

Uniqueness of the Two-Boundary Retrocausal Master Equation: a Giulini–Marolf–Aczél–Connes Theorem for the TBRME

Companion-paper architecture (rev. 2: Type II₁ gravity sector, Aczél clock re-route)

Alexandru Dima

July 15, 2026

Abstract

This companion paper proves a uniqueness theorem for the Two-Boundary Retrocausal Master Equation (TBRME) of the main unification paper. Under seven explicit axioms on a densely-defined symmetric constraint operator $\widehat{C}$ on the universal Hilbert space $\mathcal{U}$ — clock covariance with the fibrewise Aczél–d’Alembert cocycle, modular covariance with respect to the *non-tracial* dual seed state $\widehat{\omega}$ on the CLPW Type II₁ gravity sector, direct-sum locality in the branch index, TDSE flat-space limit, semiclassical-Einstein WKB limit, a pure-geometry *universality-class* condition in the sense of Kaimakkamis–Sil, and branch-uniform Part I normalization — we show $\widehat{C}$ equals $\lambda \widehat{\mathfrak{C}}_k$ uniquely up to (a) a single overall positive scale λ , (b) conjugation by unitaries of the $\widehat{\omega}$ -centralizer $\mathcal{M}_{\text{geom}}^{\widehat{\omega}}$, and (c) von Neumann self-adjoint-extension freedom where the deficiency indices of $\widehat{C}|_{\Phi}$ are non-trivial and equal. The structural template is Giulini–Marolf rigging-map uniqueness (outer frame; it also sources the per-branch scale freedom that Axiom X7 collapses), with the scalar Aczél–d’Alembert classification supplying the clock factor (Kiszyński–Sova operator-cosine theory certifying the generator), the Pedersen–Takesaki noncommutative Radon–Nikodym theorem supplying the gravity factor on the finite-trace Type II₁ sector, and Meda–Pinamonti–Siemssen semiclassical well-posedness closing the WKB leg. All external ingredients are in the ambient literature and are cited at theorem level; the contribution of this paper is assembly, an axiom independence/consistency audit, the over-determination lemma for the matter–gravity coupling, and explicit identification of the three irreducible ambiguity classes.

1 Introduction and context

Placeholder. State the main paper’s TBRME $\widehat{\mathfrak{C}}_k = \widehat{K}_k - \kappa^{-1} \widehat{Q}^{(k)}[f]$, recap the five-fold reduction theorem (U1)–(U5), and motivate the uniqueness question. Stress the “modulo centralizer” status — this is not an iff classification; it is the same uniqueness quality as the Aczél–d’Alembert classification for the clock shift.

Two corrections relative to the first scaffold are structural and propagate everywhere:

- **Type II₁, not II_∞.** With a positive clock Hamiltonian on $L^2(\mathbb{R}_+)$ and a compact region, the CLPW/DEHK crossed-product gravity sector is a Type II₁ factor with a *finite*, trace-normalized trace τ ($\tau(\mathbb{1}) = 1$) [?]; “II_∞ with induced finite trace” is a contradiction in terms. The II_∞ variant survives only on non-compact/black-hole-exterior sectors [?], treated in Rem. ??.
- **Aczél, not Kiszyński, selects the clock shape.** Per ledger item A29, $C = B + \mathbb{K}$ does not satisfy the homogeneous operator-cosine equation; the clock-sector uniqueness is the fibrewise scalar Aczél–d’Alembert classification (main paper, App. A as corrected), with Kiszyński–Sova theory certifying the reconstructed generator only. The axioms below import the corrected measurability/continuity hypotheses explicitly.

2 Standing structures and axioms

Let $\mathcal{U} = \bigoplus_{k \in \mathbb{K}_+} \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{geom}}^{(k)} \otimes \mathcal{H}_{\text{matter}}^{(k)}$, where each $\mathcal{H}_{\text{geom}}^{(k)} = L^2(\mathcal{M}_{\text{geom}}^{(k)}, \tau_k)$ carries the CLPW/DEHK crossed-product structure of App. F of the main paper in its corrected form: for a positive clock Hamiltonian on $L^2(\mathbb{R}_+)$ and compact region, $\mathcal{M}_{\text{geom}}^{(k)}$ is a *Type II₁ factor* with normalized finite trace τ_k , $\tau_k(\mathbb{1}) = 1$ [?, ?]. Let $\hat{\omega}_k = \tau_k(h_k \cdot)$ be the dual state induced by the branch seed ω_k , with Radon–Nikodym density $h_k \in L^1(\mathcal{M}_{\text{geom}}^{(k)}, \tau_k)_+$, h_k injective (faithfulness), $\tau_k(h_k) = 1$. Its modular automorphism group is *inner*,

$$\sigma_s^{\hat{\omega}_k} = \text{Ad}(h_k^{is}), \quad (1)$$

by the Pedersen–Takesaki Radon–Nikodym theorem for semifinite algebras [?] (equivalently: a von Neumann algebra is semifinite iff its modular flows are inner). In particular the modular group of the *trace* is trivial, $\sigma^{\tau_k} = \text{id}$; all modular-covariance statements below are therefore made with respect to $\hat{\omega}_k$, never τ_k .

Let $\hat{C}$ be a densely-defined *symmetric* operator on a common invariant rigged-Hilbert dense domain $\Phi \subset \mathcal{U}$ (essential self-adjointness is *not* assumed; see ambiguity (c) and Rem. ??). Assume:

Axiom 2.1 ((X1) Clock covariance and Aczél–d’Alembert cocycle). $[\hat{C}, \hat{H}_c \otimes \mathbb{1} \otimes \mathbb{1}] = 0$ on Φ , where $\hat{H}_c$ is the Page–Wootters clock generator, and the clock-shift response $B(\delta) := \hat{\Sigma}$ -part of $\hat{C}$ under finite clock shifts satisfies, on the band-limited subspace $\mathcal{H}_{[E_+]}$, the package of the main paper’s Prop. A (operator-sense d’Alembert reduction): $B(\delta)$ bounded, self-adjoint, even in δ , strongly continuous in δ on $\mathcal{D}(\hat{H}_c^2)$, with $B(\delta) \in \{\hat{H}_c\}''$ and the inhomogeneous cocycle

$$B(\delta_1 + \delta_2) + B(\delta_1 - \delta_2) = 2 \cos(\delta_1 \hat{H}_c / \hbar) B(\delta_2) + 2B(\delta_1),$$

together with $\mu_{\hat{H}_c}$ -measurability of the fibrewise data $\xi \mapsto f(\xi, \delta)$ for each rational δ .

Remark 2.2 (A29 accommodation). X1 deliberately states the *inhomogeneous* cocycle plus the measurability/continuity hypotheses under which the fibrewise scalar Aczél classification is a theorem (main paper, Prop. *wdw-dalembert* and Thm. *wdw-selection*); no homogeneous operator-cosine hypothesis on $C = B + \mathbb{1}$ is made or used. If the sibling re-route adds further measurability hypotheses, they are to be absorbed into X1 verbatim.

Axiom 2.3 ((X2) Modular covariance w.r.t. the dual seed state, non-degenerate). For each branch k there is a decomposition on Φ , $\hat{C}_k = \mathbb{1} \otimes \hat{C}_{\text{grav}}^{(k)} \otimes \mathbb{1} + \hat{C}_{\text{rest}}^{(k)}$, with $\hat{C}_{\text{grav}}^{(k)}$ symmetric and affiliated to $\mathcal{M}_{\text{geom}}^{(k)}$, $\hat{C}_{\text{rest}}^{(k)}$ commuting with h_k^{is} for all s , such that

$$\text{Ad}(e^{it\hat{C}_{\text{grav}}^{(k)}}) \upharpoonright_{\mathcal{M}_{\text{geom}}^{(k)}} = \sigma_{-\lambda_k t}^{\hat{\omega}_k} \quad \text{for some } \lambda_k > 0 \text{ and all } t \in \mathbb{R}. \quad (2)$$

Non-degeneracy: $\hat{\omega}_k$ is non-tracial, i.e. $h_k \notin \mathbb{C}\mathbb{1}$. (If h_k were scalar, $\sigma^{\hat{\omega}_k} = \text{id}$ and (??) would be vacuous; the axiom then selects nothing. Non-degeneracy holds automatically whenever the seed modular Hamiltonian K_{ω_k} is non-scalar, i.e. in every physical branch.)

Axiom 2.4 ((X3) Direct-sum locality in the branch index). $\hat{C} = \bigoplus_{k \in \mathbb{K}_+} \hat{C}_k$, with each $\hat{C}_k$ affiliated to $\mathcal{B}(\mathcal{H}_{\text{clock}}) \otimes \mathcal{M}_{\text{geom}}^{(k)} \otimes \mathcal{B}(\mathcal{H}_{\text{matter}}^{(k)})$, where $\mathcal{M}_{\text{geom}}^{(k)}$ is the Type II₁ crossed-product factor above.

Axiom 2.5 ((X4) TDSE flat-space limit, projective form). In the flat, classical-clock, finite-matter limit (reduction (U1)/(U2) of the main paper), $\hat{C}_k \downarrow c_k(\hat{H}_{\text{clock}} + \hat{H}_{\text{matter}}^{(k)})$ for some $c_k > 0$, strongly on the limit domain. (*Projective form*: only the ray of the limit generator is fixed. The strict form $c_k = 1$ would force $\lambda = 1$ in Thm. ?? and render ambiguity (a) vacuous; see Sec. ??.) The limit additionally supplies the TDSE normalization $-B''(0) = \alpha \hat{H}_c^2 / \hbar^2$ on $\mathcal{D}(\hat{H}_c^2)$ used to fix the common fibrewise scale in the clock sector.

Axiom 2.6 ((X5) Semiclassical-Einstein WKB limit). In the WKB-geometry sector, $\langle \hat{C} \rangle$ -vanishing yields the semiclassical Einstein equation $G_{\mu\nu} = (8\pi G/c^4)\langle T_{\mu\nu} \rangle$ (reduction (U3) of the main paper), within the scope where Meda–Pinamonti–Siemssen local existence–uniqueness holds [?]: *flat* FLRW backgrounds driven by a massive scalar with arbitrary curvature coupling, initial data at finite time, regular compatible state. (The earlier “compact-Cauchy FLRW” scope and the “Pinamonti–Drago–Hack” attribution were incorrect and are withdrawn.)

Axiom 2.7 ((X6) Pure-geometry universality class (weakened)). Tracing out $\mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{matter}}^{(k)}$ reduces $\hat{C}_k$ to a quantum Hamiltonian constraint whose (i) classical limit is the ADM/Wheeler–DeWitt constraint of general relativity, and whose (ii) operator-ordering ambiguity function lies in the *same Kaimakkamis–Sil universality class* [?] as the minimal (Laplace–Beltrami-covariant) ordering: under the field-redefinition-invariant reformulation, all ordering ambiguities consolidate into a single higher-order scalar function on minisuperspace, and X6 requires only membership of the minimal class. *X6 makes no reference to the cosine deformation $\hat{\Sigma}_\delta$ or to any other structure that Thm. ?? is supposed to deliver; the Σ_δ term is derived from X1+X4, not assumed here.*

Axiom 2.8 ((X7) Branch-uniform Part I normalization). On finite-dimensional matter factors, $\hat{C}_k$ restricts to the Part I two-boundary constraint (reduction (U2) of the main paper) *with a branch-independent normalization*: the constants c_k of X4 and the flow rescalings λ_k of X2 satisfy $c_k = c$, $\lambda_k = \lambda$ for all $k \in \mathbb{K}_+$. Equivalently: the Giulini–Marolf per-sector scale freedom [?] is fixed to a single global scale by Part I consistency of the branch weights $w(k)$.

Remark 2.9 (What X7 is for). Giulini–Marolf uniqueness is *per superselection sector*: on a direct sum the rigging map (and with it the normalization of the constraint) is unique only up to one positive constant per sector [?]. Under X3 every branch k is a superselection sector, so X1–X6 alone can at best give $\hat{C}_k = \lambda_k \hat{\mathcal{C}}_k$ with an uncontrolled sequence (λ_k) . X7’s entire job is to collapse $(\lambda_k) \rightarrow \lambda$; it is *not* an independent rigidity input on the operator form (that part of the old X7 was implied by X4 and has been removed — see Sec. ??).

3 Main theorem

Theorem 3.1 (TBRME uniqueness, sharp form). *Under Ax. ??–?? on the universal Hilbert space $\mathcal{U}$ of the main paper,*

$$\hat{C} = \lambda \bigoplus_{k \in \mathbb{K}_+} \hat{\mathcal{C}}_k \quad \text{for some } \lambda \in \mathbb{R}_{>0},$$

uniquely up to:

- (a) *the overall positive scalar $\lambda > 0$ (single and global, by X7; per-branch scales λ_k if X7 is dropped);*
- (b) *conjugation by unitaries of the $\hat{\omega}_k$ -centralizers $\mathcal{M}_{\text{geom}}^{(k)\hat{\omega}} = \{h_k\}' \cap \mathcal{M}_{\text{geom}}^{(k)}$ (a genuine gauge class: such conjugations fix the clock and gravity generators but act non-trivially on the gravity–matter coupling ordering);*
- (c) *a von Neumann self-adjoint-extension choice when the deficiency indices (n_+, n_-) of $\hat{C}|_\Phi$ are non-trivial and equal [?]. (Attribution corrected: this freedom is von Neumann extension theory, not Giulini–Marolf [?], whose theorem concerns rigging-map scale per superselection sector and contains no deficiency-index content.)*

Remark 3.2 (Essential self-adjointness vs. ambiguity (c)). The first scaffold assumed $\hat{C}$ essentially self-adjoint *and* kept ambiguity (c); that conjunction is inconsistent (essential self-adjointness means $(n_+, n_-) = (0, 0)$). The standing assumption is now symmetry on Φ only. In the regime where the main paper’s Kato–Rellich argument (T4)+(T7) applies, $\hat{C}|_\Phi$ is essentially self-adjoint and (c) collapses; (c) is live precisely outside that regime (e.g. at minisuperspace boundaries $a \rightarrow 0$), see Sec. ??.

4 Proof architecture

4.1 Clock sector: fibrewise Aczél–d’Alembert closure

X1 (covariance + spectral theorem) forces $B(\delta) = f(\widehat{H}_c, \delta)$; the cocycle read on each spectral fibre ξ is the *scalar* inhomogeneous d’Alembert equation, whose continuous even solutions with $f(\xi, 0) = 0$ are $f(\xi, \delta) = c(\xi)(\cos(\delta\xi/\hbar) - 1)$ (Aczél [?]); measurability of $c(\xi)$ from X1’s measurability clause; the TDSE normalization inside X4 forces $c(\xi) \equiv \alpha$. Kisyński–Sova operator-cosine theory [?, ?] enters only to certify that the normalized member is the cosine function of $A = (\widehat{H}_c/\hbar)^2 \succeq 0$. *Import from App. A of the main paper as corrected under A29; do not import the withdrawn $C = B + \mathbb{K}$ homogeneous-cosine step.* **[Status: proven]** on $\mathcal{H}_{[E_\star]}$; extension to unbounded $\widehat{H}_c$ without band-limit: **[Status: genuinely open]** (tedious spectral-limit argument, expected routine).

4.2 Gravity sector: Pedersen–Takesaki rigidity on the Type II_1 factor

The Takesaki–Connes cocycle-derivative theorem [?, Thm. VIII.3.3] is type-agnostic and survives the reclassification; but on a II_1 factor it must be deployed against the non-tracial state $\hat{\omega}_k$, since $\sigma^\tau = \text{id}$. On the finite-trace sector the argument in fact simplifies to the following elementary lemma.

Lemma 4.1 (Gravity-sector rigidity, II_1 form). *Let $\mathcal{M}$ be a II_1 factor with normalized trace τ , and $\hat{\omega} = \tau(h \cdot)$ a faithful normal non-tracial state, $h \in L^1(\mathcal{M}, \tau)_+$ injective, $\tau(h) = 1$. Suppose $\widehat{C}_{\text{grav}}$ is self-adjoint, affiliated to $\mathcal{M}$, and satisfies (??): $\text{Ad}(e^{it\widehat{C}_{\text{grav}}}) \upharpoonright_{\mathcal{M}} = \sigma_{-\lambda t}^{\hat{\omega}} = \text{Ad}(h^{-i\lambda t})$. Then*

$$\widehat{C}_{\text{grav}} = \lambda(-\log h) + \mu\mathbb{K}, \quad \mu \in \mathbb{R},$$

and $\mu = 0$ iff $\tau(e^{-\widehat{C}_{\text{grav}}/\lambda}) = 1$. In particular the II_1 trace normalization $\tau(\mathbb{K}) = 1$, $\tau(h) = 1$ canonically fixes the additive constant, a strengthening unavailable in the II_∞ setting.

Proof sketch. Set $u_t := e^{it\widehat{C}_{\text{grav}}} h^{i\lambda t}$. By hypothesis $\text{Ad}(u_t) = \text{id}$ on $\mathcal{M}$, so $u_t \in \mathcal{M}' \cap \mathcal{M} = \mathbb{C}\mathbb{K}$ ($\mathcal{M}$ a factor; $u_t \in \mathcal{M}$ since both factors are affiliated to $\mathcal{M}$). Thus $e^{it\widehat{C}_{\text{grav}}} = c(t)h^{-i\lambda t}$ with $c: \mathbb{R} \rightarrow \mathbb{T}$; strong continuity and the group law give $c(t) = e^{i\mu t}$, and Stone’s theorem yields $\widehat{C}_{\text{grav}} = -\lambda \log h + \mu\mathbb{K}$ on the common core. The normalization claim is functional calculus plus $\tau(h) = 1$. $\square$

[Status: provable now, proof to be written] (the sketch is complete modulo domain bookkeeping for the affiliated logarithm, which is standard in $L^1(\mathcal{M}, \tau)$).

Remark 4.2 (Where the centralizer ambiguity (b) actually lives). Lemma ?? pins $\widehat{C}_{\text{grav}}^{(k)}$ *exactly* (up to λ_k); conjugation by a unitary $u \in \mathcal{M}_{\text{geom}}^{(k)\hat{\omega}}$ fixes h_k (hence $\widehat{C}_{\text{grav}}^{(k)}$) and fixes $\sigma^{\hat{\omega}_k}$, but acts non-trivially on the gravity–matter coupling term $\widehat{Q}^{(k)}[f]$. Ambiguity (b) is therefore *not* a residual freedom of the gravity generator (as the first scaffold suggested); it is a residual gauge action on the *coupling ordering*. In a II_1 factor the centralizer $\{h\}' \cap \mathcal{M}$ is typically large, so (b) is a genuinely non-trivial class; partial collapse under X5 is discussed in Sec. ??, step (v).

Remark 4.3 (II_∞ variant). On non-compact sectors (black-hole exterior, CPW [?]) $\mathcal{M}$ is II_∞ with semifinite trace; Lemma ?? survives verbatim except that the density h is τ -measurable with possibly $\tau(h) = \infty$ -adjacent normalization, and the additive constant μ is no longer canonically fixed. The theorem statement is uniform: “Type II factor with faithful normal semifinite trace, finite (and normalized) in the positive-clock/compact case.”

4.3 Matter sector: the over-determination lemma

After Secs. ??–?? and X3, the residual freedom in $\hat{C}_k$ is: the pure-matter part, the gravity–matter coupling ordering in $\hat{Q}^{(k)}[f]$, and centralizer conjugation (b).

Lemma 4.4 (Over-determination of the coupling ordering). *Let $\hat{C}, \hat{C}'$ both satisfy X1–X7 and define $D_k := \hat{C}_k - \hat{C}'_k$ on Φ . Then:*

- (i) (Clock.) *The clock parts of $\hat{C}_k, \hat{C}'_k$ coincide up to the global scale; D_k commutes with the clock algebra. [Status: proven] (Sec. ??).*
- (ii) (Gravity.) *The gravity parts differ by $(\lambda - \lambda')(-\log h_k)$ only; no centralizer term survives in the generator itself. [Status: provable now, proof to be written] (Lemma ??).*
- (iii) (Pure matter.) *X4 (projective TDSE limit) forces the pure-matter parts onto the same ray: the flat-limit reduction of D_k is $(c_k - c'_k)(\hat{H}_{\text{clock}} + \hat{H}_{\text{matter}}^{(k)})$, i.e. scale only. [Status: provable now, proof to be written] (strong-limit bookkeeping on the limit domain).*
- (iv) (WKB order.) *X5 forces the leading WKB symbol of the coupling part of D_k to vanish: both constraints reduce to semiclassical Einstein flows through the same initial data, and Meda–Pinamonti–Siemssen uniqueness [?] of the semiclassical solution forces the two flows — hence the two leading symbols — to coincide, fixing in particular the coupling constant κ^{-1} relative to the matter normalization of (iii). [Status: partially proven] (MPS supply forward well-posedness as a theorem; the “equal limits + unique flow $\Rightarrow$ equal leading symbols” step is an inverse-direction argument that must be written; expected provable on the MPS scope, i.e. flat FLRW + massive scalar).*
- (v) (Ordering class.) *X6 places the pure-geometry reductions of both constraints in the Kaimakkamis–Sil minimal universality class [?]; the within-class freedom is, by their field-redefinition- invariant reformulation, a change of inner-product measure / field redefinition, i.e. unitarily implemented gauge — and the unitaries in question that preserve X2 lie in the centralizer, so the within-class freedom is absorbed into ambiguity (b). [Status: partially proven]: proven by [?] for minisuperspace models (1-dim and n-dim with scalar matter); the lift to the midisuperspace/QFT matter content of the main paper is [Status: genuinely open] (this is open problem O1, the principal conjectural input of the companion paper).*
- (vi) (Residual = extension data.) *What survives (i)–(v) is supported on minisuperspace boundary terms (e.g. $a \rightarrow 0$): symmetric, annihilated by all interior limits, hence localized in the deficiency subspaces of $\hat{C}_k \upharpoonright_{\Phi}$. This is exactly the von Neumann extension freedom, ambiguity (c). [Status: genuinely open] (conjecture O2: “interior-trivial $\Rightarrow$ boundary-supported” needs a precise boundary-triple formulation; plausible via standard boundary-triple theory on the DeWitt-metric minisuperspace, but not yet written).*

Consequently $\hat{C}_k = \lambda_k \hat{\mathcal{C}}_k$ modulo (b) and (c), with (λ_k) per-branch.

4.4 Assembly

X3 assembles the branch constraints as a direct sum. Each branch is a superselection sector; Giulini–Marolf [?] uniqueness of the rigging map holds per sector up to a positive scale, so the assembly step alone yields $(\lambda_k)_{k \in \mathbb{K}_+}$. X7 collapses $\lambda_k \equiv \lambda$, which is ambiguity (a). [Status: provable now, proof to be written] (bookkeeping once (i)–(vi) are in place).

5 Self-adjoint-extension discussion (ambiguity (c))

Spell out when deficiency indices can be non-trivial for $\hat{C} \upharpoonright_{\Phi}$: in the Kato–Rellich regime of the main paper’s (T4)+(T7) the constraint is essentially self-adjoint and (c) collapses; candidate non-trivial cases are minisuperspace boundaries ($a \rightarrow 0$) and non-compact matter sectors. Physical principles that collapse (c): positive clock spectrum, positivity of the semiclassical energy, DeWitt boundary condition. *Cite von Neumann extension theory [?, Thm. X.2] and, for the*

constraint-quantization frame, Marolf’s single-constraint RAQ [?]; the former “Giulini–Marolf case (iii)” citation is withdrawn — no such enumeration exists in gr-qc/9902045.

6 Relation to the main paper’s App. A

The clock-shift cocycle $\widehat{\Sigma}_\delta = \cos(\delta \widehat{H}_c / \hbar) - \mathbb{K}$ of the main paper’s clock-sector uniqueness (App. A, as corrected under A29: fibrewise Aczél selection, Kisiński–Sova certification) is the clock-sector projection of Thm. ?? . The present paper extends that closure from the clock factor to the full universal Hilbert space $\mathcal{U}$.

7 Axiom audit: dependency, independence, consistency

Dependency diagram

X1 (cocycle + measurability)	$\xrightarrow{\text{Aczél fibrewise}}$	clock shape $\alpha \widehat{\Sigma}_\delta$
X4 (TDSE, projective)	$\xrightarrow{\text{normalization}}$	fibre scale $c(\xi) \equiv \alpha$; pure-matter ray
X2 ($\widehat{\omega}$, non-tracial)	$\xrightarrow{\text{Pedersen–Takesaki}}$	$\widehat{C}_{\text{grav}} = \lambda(-\log h)$ exactly
X5 (WKB + MPS)	$\xrightarrow{\text{unique s.c. flow}}$	leading coupling symbol, κ
X6 (KS class)	$\xrightarrow{\text{within-class} = \text{gauge}}$	coupling ordering mod (b)
X3 (direct sum)	$\xrightarrow{\text{GM per-sector}}$	assembly with (λ_k)
X7 (branch-uniform)	$\longrightarrow$	$\lambda_k \equiv \lambda$ (ambiguity (a))
residual of (i)–(v)	$\xrightarrow{\text{boundary triples}}$	extension data (ambiguity (c))

Joint use: clock closure = $X1 \wedge X4$; matter closure = $X4 \wedge X5 \wedge X6$ (the over-determination); gravity closure = X2 alone; assembly = $X3 \wedge X7$.

Independence and non-redundancy

- **X4 vs X7 (redundancy found and removed).** The old X7 (“restricts to the Part I constraint on finite-dim factors”) was, at the level of the constraint *operator*, implied by X4: the flat classical-clock limit through finite-dimensional matter factors already fixes the Part I generator. The non-redundant content of X7 is only the branch-*uniform* normalization (and the boundary-data/Born-weight structure, which concerns the state space, not $\widehat{C}$, and is moved out of the axiom set). X7 has been restated accordingly.
- **X4 strict form would kill ambiguity (a).** If X4 demanded $c_k = 1$ (exact reduction), then $\lambda = 1$ and the “up to scale” clause of Thm. ?? is vacuous. The projective form is the consistent choice.
- **X6 conclusion-smuggling (referee attack sustained, axiom weakened).** The old X6 required reduction to “the modified Wheeler–DeWitt constraint of (U4)”, whose statement contains $\alpha \widehat{\Sigma}_\delta$ — i.e. it assumed on the clock–geometry cross-sector precisely what X1+X4 are supposed to derive. The weakened X6 above is a pure universality-class membership statement (Kaimakkamis–Sil), makes no reference to Σ_δ , and demonstrably does independent work only in step (v) of Lemma ??.
- **X2 near-smuggling (assessed, kept, scoped).** X2 identifies the gravity generator’s flow with the $\widehat{\omega}$ -modular flow; this is the main paper’s thermal-time postulate, and relative to it X2 does the bulk of the gravity-sector work (Lemma ?? is then short). This is legitimate *relative* uniqueness (as Poincaré covariance is for P^μ), but the theorem must be advertised as “unique given thermal time,” not as deriving thermal time. The non-degeneracy clause ($h \notin \mathbb{C}\mathbb{K}$) is required: without it X2 is vacuous on the Π_1 sector and the conjunction X1–X7

loses gravity-sector selectivity entirely — this is the new failure mode introduced by the II_1 reclassification, and it is exactly why X2 cannot be stated w.r.t. the trace.

- **Mutual consistency / non-vacuous conjunction.** The de Sitter static-patch witness model of the main paper’s (H6)/(U6) satisfies X1–X7: it is *natively* II_1 [?], so the reclassification improves rather than damages non-vacuity. $\widehat{C} = \widehat{\mathfrak{C}}_k$ itself satisfies all seven axioms on that model by construction.
- **X1’s old form had a vacuous clause.** The commutator condition “ $[\widehat{C}, \widehat{\Pi}_c + \mathbb{K} \otimes \mathbb{K}] = 0$ ” contained an identity summand that commutes with everything; removed.

8 Open questions and triage

Provable now (write-up work only)

- Clock closure on $\mathcal{H}_{[E_\star]}$ (done in the main paper post-A29; import).
- Gravity rigidity Lemma ?? (elementary on II_1 ; Pedersen–Takesaki [?] + factor triviality of the relative commutant).
- Over-determination steps (i)–(iii); assembly + X7 scale-tying; the audit of Sec. ??.

Partially proven (external theorem in place, inverse/bridging argument to write)

- Step (iv): MPS-based WKB rigidity (forward well-posedness is [?]; inverse direction to write; scope limited to flat FLRW + massive scalar).
- Step (v) on minisuperspace ([?] is the input theorem; the “within-class = centralizer gauge” identification to write).

Genuinely open

- O1** Lift of the Kaimakkamis–Sil universality classification beyond minisuperspace to the midisuperspace/QFT matter content of the main paper. *Principal conjectural input.*
- O2** Step (vi): interior-trivial residuals are boundary-supported (boundary-triple formulation on DeWitt-metric minisuperspace).
- O3** Partial collapse of the centralizer class (b) by X5 beyond leading WKB order.
- O4** Clock closure for unbounded $\widehat{H}_c$ without the band-limit.
- O5** Relaxing X3 to a continuous field of II_1 factors over the branch index.
- O6** Non-perturbative dynamical-horizon regime (II_∞ sectors of Rem. ??; residual (E3a) of the main paper).

References

- [1] A. Dima, *A Two-Boundary Retrocausal Unification of Quantum Mechanics and General Relativity: Rigorous Core and a Conditional Master Theorem* (main paper; 2026).
- [2] D. Giulini, D. Marolf, *A uniqueness theorem for constraint quantization*, Class. Quantum Grav. **16**, 2479 (1999), arXiv:gr-qc/9902045 [verified: rigging map unique up to scale per superselection sector under absolutely convergent group averaging; no deficiency-index content]; D. Marolf, *Refined algebraic quantization: systems with a single constraint*, arXiv:gr-qc/9508015.
- [3] J. Kiszyński, *On cosine operator functions and one-parameter groups of operators*, Studia Math. **44**, 93–105 (1972) [verified].
- [4] M. Sova, *Cosine operator functions*, Rozprawy Mat. **49**, 1–47 (1966).

- [5] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press (1966).
- [6] M. Takesaki, *Theory of Operator Algebras II*, Springer (2003). [Connes cocycle-derivative theorem: Thm. VIII.3.3; type-agnostic. The converse characterization of unitary σ -cocycles as derivatives is in §VIII.3; exact numbering to be checked against the 2003 printing.]
- [7] G. K. Pedersen, M. Takesaki, *The Radon–Nikodym theorem for von Neumann algebras*, Acta Math. **130**, 53–87 (1973) [verified; includes: $\mathcal{M}$ semifinite iff modular automorphism groups are inner].
- [8] P. Meda, N. Pinamonti, D. Siemssen, *Existence and uniqueness of solutions of the semiclassical Einstein equation in cosmological models*, Ann. Henri Poincaré **22**, 3965 (2021), arXiv:2007.14665 [verified: flat FLRW, massive scalar with arbitrary curvature coupling, local existence–uniqueness via Banach fixed point].
- [9] E. Kaimakkamis, K. Sil, *A formalism for the ambiguities of the Wheeler–DeWitt equation*, arXiv:2407.03077 (2024) [author attribution corrected from the earlier erroneous “L. Chataignier et al.”; verified: operator-ordering ambiguities consolidate into a single higher-order scalar function on minisuperspace, yielding universality classes with order-by-order universal semiclassical predictions].
- [10] V. Chandrasekaran, R. Longo, G. Penington, E. Witten, *An algebra of observables for de Sitter space*, JHEP **02** (2023) 082, arXiv:2206.10780 [verified: the static-patch observer algebra is Type II₁].
- [11] V. Chandrasekaran, G. Penington, E. Witten, *Large N algebras and generalized entropy*, JHEP **04** (2023) 009, arXiv:2209.10454 [black-hole exterior: Type II_∞].
- [12] M. Reed, B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*, Academic Press (1975) [von Neumann deficiency-index theory, Thm. X.2].